Abhishek H S

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Area of interest :AI&ML,DevOps

Education

- **PES University (7th sem)** Bangalore, India
  Bachelor of Technology in Computer Science(AI-ML) *Nov. 2022 – Present*

Skills

Languages: Python, C, Javascript, Bash

Libraries/Frameworks:OpenCV,Tensorflow,Pandas,Numpy,Scikit-Learn,Matplotlib,NLTK,

Tools:Docker,Git,Fast API,Linux,AWS,Transformers,Diffusion,LLM's,GenAI,Langchain,RAG

Experience

- **Appex PESU** PES University, Bangalore
  Open Source Contributor
  - **4D Cardiac Dynamics Modeling and Visualization for Surgical Innovation** ***(source code)***
    - Developed a 4D heart model to analyze cardiac motion and tissue deformation, enhancing cardiovascular research.
    - Integrated autoencoders and CycleGAN architectures to enhance simulation realism

- **Appex PESU** PES University, Bangalore
  Logistics head *Jan 2024 - Present*
  - **App Genesis**:
    - Organized and led an 24-hour ideathon called Horcrux by AppexClub at PES University, focusing on technical aspects such as mentoring participants on development, providing guidance on coding challenges, and evaluating innovative solutions.

- *IOT PESU* PES University, Bangalore
  Contributor *Dec2025 -Present*
  - **Chat-Bot (source code)**
    - Developed a FastAPI-based healthcare chatbot using RAG with Gemini API to enable context-aware conversations
    - Implemented intelligent document processing with OCR , supporting PDF/image extraction

Projects

- Word2Vec: **(source code)**
  - Implemented Word2Vec (CBOW and Skip-gram) using Python and NumPy.
  - Trained custom word embeddings on large text corpus, achieving meaningful semantic relationships and vector analogies.

- Plant Disease Prediction: **(source code)**
  - Built a neural network in TensorFlow/Keras for plant disease detection from images.
  - Enhanced model accuracy by implementing convolutional neural networks (CNNs) alongside advanced techniques like data augmentation and dropout. These methods improved generalization and reduced overfitting.

- Handwriting extraction and translation: **(source code)**
  - Built service using Google Gemini Vision API for OCR, handwriting recognition, and medical data extraction from prescription
  - Developed automated translation and medical terminology extraction system supporting 10+ languages.

Actively participated in multiple intercollegiate hackathons, gaining hands-on experience in real-time problem-solving, cross-functional teamwork, and rapid prototyping using modern development tools.
Certification: View certificates